

BP EXPLORATION OPERATING COMPANY LTD
North Sea Decommissioning — Forties Pipeline System
REQUEST FOR QUOTATION — CENTRIFUGAL PUMP PACKAGES

Doc No: BP-RFQ-2024-FPS-0112 | Rev: A | Date: 2024-10-01

Project	Forties Pipeline System — Decommissioning Phase 2
Client	BP Exploration Operating Company Ltd
Commodity	API 610 Centrifugal Pump Packages (OH2, BB3 types)
Quantity	18 pump packages (see MTO attached)
Bid Deadline	2024-11-15 14:00 GMT
Delivery	2025-09-01 — Aberdeen Quayside
Incoterms	DDP Aberdeen
Currency	GBP

1. SCOPE OF SUPPLY

1.1 This RFQ covers the supply of 18 centrifugal pump packages for crude oil transfer service on the Forties Pipeline System decommissioning project. All pumps shall be suitable for continuous offshore duty in a Zone 1 classified hazardous area.

Tag	Type	Duty	Flow (m3/h)	Head (m)	Qty
10-P-001/002	OH2 (overhung, between bearing)	Crude transfer primary	850	380	4
10-P-003/004	OH2	Crude transfer secondary	420	220	4
10-P-005/008	BB3 (between bearing, double suction)	Pipeline booster	2400	560	4
10-P-009/012	BB3	Seawater injection	1800	720	4
10-P-013/018	OH2	Chemical dosing	45	180	2

2. MATERIAL REQUIREMENTS

2.1 Casing & Cover

Pump casings shall be ASTM A216 Grade WCB carbon steel for crude service. Seawater injection pumps (10-P-009 to 012) SHALL use Super Duplex 2507 (UNS S32750) casings. Note: Section 6.3 of this specification references carbon steel casings for all pump types — vendors must clarify which material applies to seawater pumps.

2.2 Impeller

Impellers for crude service: ASTM A351 CF8M (316 SS). Seawater service impellers: Super Duplex 2507. Impeller trim shall not exceed 95% of maximum diameter. Closed impellers are mandatory for all BB3 type pumps.

2.3 Shaft

Pump shafts shall be AISI 4140 alloy steel, heat treated to minimum 28 HRC. Maximum shaft deflection at seal faces shall not exceed 0.05 mm under any operating condition. Duplex stainless steel shafts (UNS S31803) are required for seawater injection service.

2.4 Mechanical Seal

Mechanical seals shall be Plan 11 (API 682) as minimum for crude service. Seawater injection pumps require Plan 32 (external flush) with a dedicated clean water supply. Seal faces: Silicon Carbide vs Silicon Carbide for abrasive service, Carbon vs Silicon Carbide for clean service.

3. APPLICABLE STANDARDS

Standard	Description	Applicability
API 610 (12th Ed)	Centrifugal Pumps for Petroleum Industry	All pumps — mandatory
API 682 (4th Ed)	Pumps — Shaft Sealing Systems	Mechanical seals — mandatory
API 670	Machinery Protection Systems	All pumps >75 kW
API 671	Special Purpose Couplings	BB3 pumps only
ATEX Directive 2014/34/EU	Explosive Atmospheres Equipment	Zone 1 — Ex d IIB T3
NACE MR0175	Sour Service Materials	If H2S detected in crude
ISO 13709	Centrifugal Pumps — identical to API 610	Alternative to API 610
IEC 60034	Rotating Electrical Machines	Drive motors
BP GIS 32-301	BP Group Instruction for Rotating Equipment	Supersedes where more stringent

3.1 ISO 13709 is technically identical to API 610 — vendors offering ISO 13709 compliance shall confirm equivalence in writing. Note: Section 7.2 mandates API 610 11th Edition, while this cover sheet references the 12th Edition — the 12th Edition shall take precedence.

4. TESTING REQUIREMENTS

Test	Standard	Acceptance Criteria	Witness
Hydraulic performance test	API 610 §8.3.2	Grade 1B tolerance	BP TPA mandatory
NPSH test	API 610 §8.3.3	3% head drop method	BP TPA mandatory
Mechanical running test (4hr)	API 610 §8.3.4	Vibration <4.5 mm/s RMS	BP TPA mandatory
Hydrostatic pressure test (1.5× MAWP)	API 610 §8.2	No leakage 30 min	Cert review
Complete unit test with motor	API 610 §8.3.6	Power within 3% predicted	BP TPA mandatory
Vibration analysis (FFT)	API 670	Sub-synchronous <25% of BEP flow	BP TPA mandatory
Noise level test	ISO 3744	Max 85 dB(A) at 1m	Cert review
ATEX verification	IEC 60079-0	Zone 1 Ex d IIB T3 Gc	Notified body

4.1 Vibration acceptance: API 610 Table 9 limits apply. However BP GIS 32-301 Section 4.6 specifies a stricter limit of 3.5 mm/s RMS for offshore duty — the BP GIS limit shall govern. Vendors to confirm test rig capability.

5. COMMERCIAL TERMS

-
- 5.1 Payment: 25% on order, 65% on FAT completion, 10% on delivery. Net 45 days from invoice.
 - 5.2 Warranty: 24 months from commissioning or 30 months from delivery, whichever is earlier.
 - 5.3 Liquidated Damages: 0.75% per week of delay, capped at 7.5% of contract value.
 - 5.4 Lead time: Maximum 20 weeks. Bids exceeding 24 weeks delivery shall be rejected.
 - 5.5 Spare parts: 2-year operational spares list mandatory. Commissioning spares included in base price.

6. SPECIAL TECHNICAL REQUIREMENTS

- 6.1 All pumps shall be supplied with API 610 12th Edition baseplate (Type A — grouted). Exception: chemical dosing pumps (10-P-013/018) may use Type C baseplates per vendor standard.
- 6.2 Coupling guard shall be fabricated from 316 SS. Non-sparking material required per ATEX Zone 1. Note: Section 3 specifies IIB gas group; offshore survey indicates IIC classification may apply — vendor to confirm gas group with BP hazardous area engineer before finalising.
- 6.3 Casings for all pump types shall be carbon steel per ASTM A216 WCB. [Note: contradicts Section 2.1 which mandates Super Duplex for seawater pumps — Super Duplex takes precedence for 10-P-009 to 012].
- 6.4 Driver: All pumps shall be driven by ATEX-certified electric motors (Ex d IIB T3). Variable speed drives (VFDs) are mandatory for all BB3 type pumps. VFD enclosure rating: IP55 minimum.